MATLAB simulation codes for [1, Section 4]

1. Overview

This MATLAB code simulates pedestrian (crowd) dynamics and infection propagation in a confined domain with a single exit using a second-order macroscopic model from [1, Sect 4].

It couples:

- A **hydrodynamic model** for pedestrian density and velocity (Roe's scheme with MUSCL reconstruction);
- A **drift–diffusion–reaction model** for airborne infection potential (Rusanov flux);
- A **SEIS compartmental model** for infection spread (susceptible–exposed–infected–protected).

The simulation outputs real-time contour maps and a video showing the evolution of:

1. Total pedestrian density ρ,
2. Infection coefficient β_I,
3. Exposed population density ρ^E.

2. Key Numerical Methods

- **Crowd Flow Solver:** Roe's approximate Riemann solver for the second-order macroscopic flow equations [3].
- **Flux Reconstruction:** Monotone Upstream-Centered Scheme for Conservation Laws (MUSCL) with flux limiter function [4].
- **Eikonal Equation:** Solved via the Fast Sweeping Method (FSM) from [2], defining pedestrian desired directions.
- **Infection Potential β:** Solved by Rusanov's scheme with MUSCL reconstruction for the advection part and second-order finite differences for the diffusion part.
- **Epidemic Dynamics:** SEIS model coupled with pedestrian advection.

3. File Structure

`Main_Crowd_SEIS.m`	Main driver file
`Approx_Roe_PW.m`	Roe solver for flux computation
`lim.m`	Flux limiter function
`FSG.m`	Fast sweeping solver for the Eikonal equation
`UV.m`	Desired speed model based on local density
`potential.m`	For ventilation potential solver for airflow
`RESULTS.m`	Text file with time, total pedestrians, and exposed counts
`Test_Case.mp4`	Output movie showing crowd and infection evolution

4. Simulation Setup Example Problem

- **Domain:** $10\,\mathrm{m} \times 10\,\mathrm{m}$ rectangular room.
- **Exit:** $2\,\mathrm{m}$ wide, centered on the right boundary.
- **Grid spacing:** $\Delta x = \Delta y = 0.1\,\mathrm{m}$.
- **Time step:** $\Delta t = 0.01\,\mathrm{s}$.
- **Total time:** 3000 iterations ($\approx 30\,\mathrm{s}$ simulation).

Initial pedestrian density is prescribed in a rectangular region $1 \leq x \leq 5$, $2.5 \leq y \leq 7.5$ with $\rho_0 = 2.4$ pedestrians/m^2.

5. Model Parameters

Symbol	Meaning	Default Value
τ	Relaxation time	0.6 s
v_f	Free flow speed	1.4 m/s
R_m	Maximum density	6 ped/m^2
S	Diffusion coefficient (infection)	1.2×10^{-3}
i_0	Infection rate	0.04
ϵ_1	Numerical threshold	10^{-4}
lose	Particle loss coefficient	0.5
kap	MUSCL reconstruction parameter	0

6. Output Files

- `Test_Case.mp4`: Movie file showing temporal evolution of crowd density, infection coefficient, and exposed density.
- `RESULTS.m`: Text file containing:

  time (s) total exposed pedestrians total pedestrians in domain.

7. Running the Simulation(s)

1. Ensure all required MATLAB functions (`Approx_Roe_PW.m`, `FSG.m`, etc.) are in the same directory.
2. (Optional) If including ventilation, run:
   ```
   potential
   ```
   and comment out the first line in `Main_Crowd_SEIS.m` that clears the workspace as well as `ug=zeros(Ny,Nx); vg=zeros(Ny,Nx)` in lines 390-391.
3. Run the main code:
   ```
   Main_Crowd_SEIS
   ```
4. View progress in MATLAB figures and the generated movie.

8. Notes and Recommendations

- Adjust Δt, Δx, and τ to satisfy the Courant–Friedrichs–Lewy (CFL) condition to satisfy eq. (29) in [1]
- Modify the initial infection rate i_0 or diffusion coefficient S to simulate various contagion scenarios.
- For ventilation studies, activate airflow fields computed by `potential.m`.
- Visualization parameters (color limits, view angles) can be modified in the plotting section.

9. References

1. A.I. Delis, N. Bekiaris-Liberis, "Numerical investigation of the effect of macro control measures on epidemic transport via a coupled PDE crowd flow - epidemic spreading dynamics model", *Applied Mathematics and Computation*, Volume 513, 2026.
2. E. Zhao, "Fast sweeping methods for Eikonal equations," *Math. Comput.*, 2005.
3. Roe, P. L., "Approximate Riemann solvers, parameter vectors, and difference schemes", *J. Comput. Phys.*, 1981.
4. E.F. Toro, Riemann Solvers and Numerical Methods for Fluid Dynamics, Springer Berlin, Heidelberg, Heidelberg, 2009.